

AMS 518 – Fall 2025 Project Review

“Bivariate Conditional Normal Distribution – Parameter Calibration”

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Reviewer:

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Executive summary

Anderson implements a directional-quantile calibration method for a conditional bivariate Gaussian model whose mean and covariance depend linearly on a scalar factor. The main numerical contribution is a controlled synthetic experiment showing that a full joint grid search over the five unknown parameters can recover the data-generating values; the report documents failed attempts with a PSG-based formulation and a sequential calibration strategy before settling on the grid-search approach. Diagnostics (one-dimensional slices and two-dimensional contours) are used to illustrate identifiability and the shape of the Koenker–Bassett (KB) loss surface. Overall the project is well-focused, technically sound for the synthetic setting, and pedagogically useful as a demonstration of the directional quantile framework. The remainder of this review follows the requested structure and assigns section scores with brief rationale and actionable suggestions.

1 Complexity and clarity of problem / model formulation (0–40 pts)

(Assessment: **39/40**)

What is formulated? The report specifies a conditional bivariate normal model

$$Y \mid X \sim \mathcal{N}_2(\mu(X), \Sigma(X))$$

with mean and covariance linearly dependent on a scalar factor X . Unknown parameters are the two mean slopes β_{11}, β_{21} and the three covariance slopes $\sigma_{11}, \sigma_{12}, \sigma_{22}$. The directional quantile functional and the Koenker–Bassett error are carefully defined and used to construct the empirical loss that is minimized.

Deterministic vs stochastic; stage-setting. The model is stochastic (conditional Gaussian) but it is a *single-stage* conditional model (no multi-stage dynamics). The report makes this clear: conditioning is on a contemporaneous factor X , not on a decision tree or time-evolving filtration.

Measure of loss / risk and constraints. The objective is the empirical directional KB loss (quantile loss aggregated over directions a , quantile levels α , and observations). Positivity constraints

on directional variances are included. The meaning of the performance function (KB loss as a quantile-based discrepancy measure) is explained and motivated via the risk-quadrangle literature. This is appropriate and well argued. It's actually a very smart choice to convert this minimization problem

$$\min_{\theta} \int_A \int_{S^1} w(a, \alpha) \mathcal{E}_{\alpha}[L(a, \alpha, X; \theta)] da d\alpha,$$

into the following

$$\min_{\theta} \sum_{k=1}^K \sum_{j=1}^J w(a^{(k)}, \alpha_j) \mathcal{E}_{\alpha_j}[L(a^{(k)}, \alpha_j, X; \theta)],$$

via a discretization over unit directions and quantile levels.

Advantages and drawbacks of the formulation.

- *Advantages:* The directional projection reduces a multivariate estimation problem to a collection of scalar quantile problems; closed-form directional quantiles under Gaussianity make the objective explicit; the approach yields interpretable diagnostics (slices/contours).
- *Drawbacks / limitations:* the covariance parameters enter through a square-root scale term, producing a nonlinear dependence that complicates direct convex optimization (PSG) and forces the author to use a brute-force grid search. The current formulation does not report computational cost or scalability metrics.

Recommendation to improve this section: Include a short paragraph on expected scaling (how K, J, and N affect cost) so the reader can judge feasibility for larger problems.

2 Adequacy and correctness of approaches (0–40 pts)

(Assessment: 38/40)

Methods used: The author attempts (i) a direct PSG-based optimization, (ii) a sequential mean/covariance calibration, and (iii) a full joint grid search. These are carefully documented; the PSG and sequential approaches are analyzed and their failings explained, leading to the final grid-search implementation.

Correctness: For the synthetic experiment the methods are implemented correctly: the KB-loss is computed as specified, vectorized evaluations are used for efficiency, and diagnostics show the grid-search minimizer coincides with the true parameter vector. Reporting of slice and contour plots supports the identifiability claims. The auxiliary-variable PSG reformulation is mathematically valid but impractical at the sample sizes used; the author correctly recognizes dimensionality and design-matrix issues.

Adequacy and alternatives: While the grid search is a defensible and transparent choice for a synthetic verification experiment, its scalability is limited.

Recommendation to improve this section: Add a short experiment comparing the full grid with a coarser-to-finer nested search (or Bayesian optimization) to show how quickly the minimizer is approached; report runtimes and memory footprints. Also, it could be considered what is the effect of choosing other parameters used for generating benchmark data.

3 Presentation style, clarity of graphs and tables (0–20 pts)

(Assessment: **20/20**)

Strengths:

- The report is well structured (introduction, model, calibration problem, numerical implementation, results, conclusion).
- Figures (one-dimensional slices and two-dimensional contours) are appropriate and effectively communicate the shape of the KB loss surface and identifiability.
- The paper explains very well why PSG failures occurred and the rationale for each method attempted.

Weaknesses / suggestions:

- The report omits computational resource reporting (time / memory), which is essential when advocating a grid-search strategy.

4. Reviewer’s conclusion: positives, negatives and section grades

Positive aspects

- This is a solid and pedagogically valuable project that clearly demonstrates the directional-quantile calibration idea.
- Clear and rigorous formulation of the directional quantile calibration objective and consistent notation.
- Thorough diagnostic analysis (slices, contours) that convincingly demonstrates identifiability in the synthetic experiment.
- Honest and useful documentation of failed approaches (PSG, sequential) which provides pedagogical value for others facing similar modelling constraints.

Limitations

- Scalability and computational-cost analysis is missing. The use of a dense 9-point-per-parameter grid (59,049 evaluations) may be acceptable for didactic synthetic work but needs justification for larger models.

Section	Max points	Awarded
1. Complexity & clarity of formulation	40	39
2. Adequacy & correctness of methods	40	38
3. Presentation style, graphs, suggestions	20	20
Total	100	97

Table 1: Section grades (reviewer).

End of review.